60-MINUTE RAG LESSON PLAN

When RAG Finds the Right Document—but Not the Answer

Audience: Developers and solution architects with basic Python | **Format:** Explain → demonstrate → practise → diagnose → improve | **Facilitator:** Nicodemus Chan

INDEPENDENT TEACHING SAMPLE: This lesson plan is an independent demonstration by Nicodemus Chan; it is not official OpenAI curriculum.

OPENING INCIDENT (0–5 min): Before revealing the result, ask: "Will the system retrieve, abstain, or invent an answer?" Then show the frozen case: the meal-allowance question scores 0.49 against a 0.45 threshold and retrieves travel_policy.txt—but the policy contains no maximum daily amount.

Five measurable learning outcomes

1. Draw and explain the ingestion and query paths, identifying where ranking, filtering, and generation occur.
2. Predict retrieve/abstain outcomes from a question, top score, and frozen threshold, then justify the prediction.
3. Calculate TP, FP, TN, FN, precision, and recall from the six held-out cases (3, 1, 2, 0 → 75% precision; 100% recall).
4. Diagnose a retrieval trace as relevance failure, threshold failure, evidence-sufficiency failure, or generation failure.
5. Recommend one evidence-based improvement without test leakage: expand labeled cases, add a sufficiency check, or recalibrate on validation data.

Session sequence and formative checks

TIME	STAGE	FACILITATOR + PARTICIPANT ACTIVITY	FORMATIVE KNOWLEDGE CHECK
0–5	**Opening / predict**	Commit to retrieve, abstain, or answer before seeing the result.	Cold-call: What evidence would change your prediction?
5–15	**Explain**	Use slides 2–5: ingestion vs query; cosine ranking; frozen threshold; relevance ≠ sufficiency; held-out metrics.	Point to where 0.45 acts. Is 0.49 proof that the requested fact exists?
15–25	**Demonstrate**	Run rag_app.py evaluate; trace the meal-allowance row from score → source → chunk text → expected abstention.	Learners identify TP/FP/TN/FN and calculate 75% precision, 100% recall.
25–40	**Participant practice**	Pairs diagnose three traces using: question → score → threshold → source → evidence text → expected behavior.	Require a failure label, evidence quote, and one justified next action.
40–50	**Debrief / improve**	Compare threshold tuning, labeled-set expansion, sufficiency checks, and validation recalibration.	Vote: Which change avoids peeking at the held-out test—and why?
50–56	**Agent demonstration**	Run rag_agent.py with the same question. Show that the agent calls the existing RAG tool and says the amount is not specified.	What decision belongs to retrieval, and what decision belongs to the agent?
56–60	**Assessment / exit**	Individual written response to the exit question below; sample two answers aloud.	Success requires correct action, diagnosis, and non-leaking improvement.

Retrieval-failure diagnostic exercise

PAIR TASK: For each trace, classify the failure, cite the decisive evidence, and recommend the smallest defensible change. Required trace: meal allowance → 0.49 → travel_policy.txt → meals are reimbursable with receipts, but no maximum is stated. Expected diagnosis: relevant retrieval, insufficient evidence; answer should abstain rather than invent.

MATERIALS + PREREQUISITES: Slides; PyCharm or terminal; Python environment and project files; knowledge_base/*.txt; rag_index.json; rag_app.py; rag_agent.py; API key and network access for live calls. Learners need basic Python and a plain-language introduction to embeddings/cosine similarity. Prepare saved console output and the three policy files for an offline fallback.

COMMON PROBLEMS + FALLBACKS: Missing key, quota, or network → use frozen console output. Missing/stale index → run rag_app.py ingest; if unavailable, provide the retrieved chunk. Scores differ after model/corpus change → teach from the frozen 0.49/0.45 trace and discuss reproducibility. Agent SDK/model error → demonstrate retrieval plus the saved grounded response. Wrong working directory → run from the project root. A missing meal limit is the intended edge case, not a broken demo.

FINAL EXIT QUESTION: "A result scores 0.49 against a frozen 0.45 threshold and retrieves the relevant travel policy, but the requested amount is absent. What should the system do, what failed, and what would you improve without leaking the held-out test?" Expected: retrieve then abstain; diagnose evidence insufficiency; add labeled development cases and a sufficiency check, then recalibrate only on validation data.